Predicting Total Medicare Beneficiaries by State (2023)

By: Team G11 (Sidharth Dusanapudi, Ashish Sunkarapalli,
Jayden Sincere Brown, Krish Venkatram Mallina)

Introduction

1. Do states with certain combinations of average age and political affiliation have higher total Medicare beneficiary counts?

2. Do states with greater healthcare provider availability (measured by physicians per 100,000 residents) have higher total Medicare beneficiary counts?
3. Do states with higher median family incomes have lower total Medicare beneficiary counts?

Data Summary

Metadata Summary

Intentionally left blank for now.

EDA Summary

This report only uses one 2023 observation for each of the 50 states. Before calculating the population quartiles, the total-state-population values for AK, AL, AR, and AZ were adjusted to be consistent with the listed state abbreviations provided in the original data set. The response variable, total Medicare beneficiaries, is strongly right-skewed: counts range between 120,280 and 7,080,940, with mean of 1,360,708, a median of 994,359, and IQR of 1,132,502. Because the mean is well above the median and the strong influence of a small number of extremely large states dominating the upper tail, a log transformation of the response variable seems appropriate before fitting the final regression model.

Among the quantitative predictors, HCC score has the clearest positive association with beneficiary totals, with Pearson correlation 0.562 on the untransformed response variable and 0.702 when using `log(Y)`. Percent male shows the strongest negative correlation of the quantitative predictors, with correlations -0.28 on the untransformed response variable and -0.467 when using `log(Y)`. Average age appears weak on its own (r = -0.038 on the raw scale), which makes the political affiliation-age interaction worth investigating rather than relying on a simple main-effect interpretation. Median income is nearly unrelated to the raw response variable (r = 0.051) and only very weakly negatively correlated when using `log(Y)` (r = -0.019), so any income effect may depend on adjustment for other predictors.

In terms of qualitative predictors, population quartile exhibits the strongest visual separation, where the fourth-quartile average is at 2,360,061 whereas in the first quartile it is only

860,492. Political affiliation also shows some distinctions in means of the number of beneficiaries, which for Swing and Democratic states is higher compared to Republican states. On the other hand, the relationship between beneficiary totals and categorical predictors such as education and physician quartile is somewhat less consistent, as the average number of beneficiaries increases from Q1 (which is 819,413) to Q3 (2,083,766) before declining in Q4.

When assessing potential interactions using log(Y), the relationship between average beneficiary age and log-transformed beneficiaries varies heavily depending on political affiliation. While Republican and swing states exhibit relatively flat trends, Democratic states display a steep negative association with log(Y), pointing to a strong interaction effect that warrants inclusion in the final model. Other investigated interactions, including HCC score by physician quartile and median income by population quartile plotted against log(Y), yielded nearly parallel trend lines, indicating no substantial interactive effects among those specific explanatory variables.

Analysis and Results

FILL THIS SECTION WITH ANALYSIS OF PLOTS PRESENTED IN APPENDIX D

Conclusions

Appendix A: Data Dictionary

Data dictionary for the cleaned analysis dataset.

Variable	Type	Description
State	Identifier	Two-letter state abbreviation.
Beneficiaries	Response	Total Medicare beneficiaries in the state (measured in people).
Avg_Age	Quantitative	Average beneficiary age (measured in years).
HCC_Score	Quantitative	Hierarchical condition category risk score (unitless index/multiplier).
Pct_Male	Quantitative	Proportion of beneficiaries who are male (measured from 0 to 1).
Median_Income	Quantitative	Median annual family income (measured in US Dollars).
Political_Affiliation	Qualitative	State political affiliation. Levels: Democrat, Republican, Swing.
Education_Q	Qualitative	Quartile of bachelor's degree percentage by state. Levels: Q1 Low, Q2, Q3, Q4 High.
Physician_Q	Qualitative	Quartile of physicians per 100,000 residents. Levels: Q1 Low, Q2, Q3, Q4 High.
Population_Q	Qualitative	Quartile of corrected total state population. Levels: Q1 Low, Q2, Q3, Q4 High.

Appendix B: Data Rows

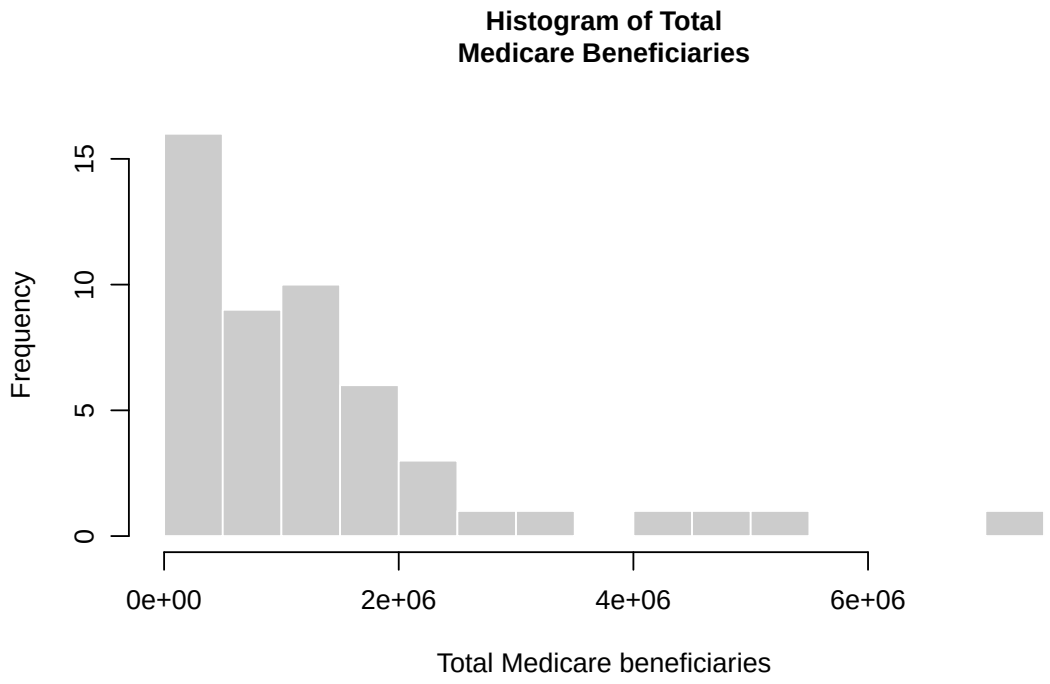
`head()` of the cleaned analysis data.

```
head(analysis_data)
```

```
  State Beneficiaries Avg_Age HCC_Score Pct_Male Median_Income
1    AK        120280    36.5      0.82   0.4918        105752
2    AL       1151244    39.6      1.02   0.4535         79940
3    AR        699823    38.9      1.00   0.4531         74179
4    AZ       1528056    39.3      0.92   0.4635         93553
5    CA       7080940    38.2      1.04   0.4567        109349
6    CO       1055744    37.9      0.89   0.4708        115335
  Political_Affiliation Education_Q Physician_Q Population_Q log_Beneficiaries
1            Republican          Q2          Q3       Q1 Low          11.69758
2            Republican      Q1 Low      Q1 Low           Q3          13.95635
3            Republican      Q1 Low      Q1 Low           Q2          13.45858
4            Republican          Q2          Q2           Q3          14.23951
5              Democrat          Q3          Q3      Q4 High          15.77292
6                 Swing     Q4 High          Q3           Q3          13.86976
```

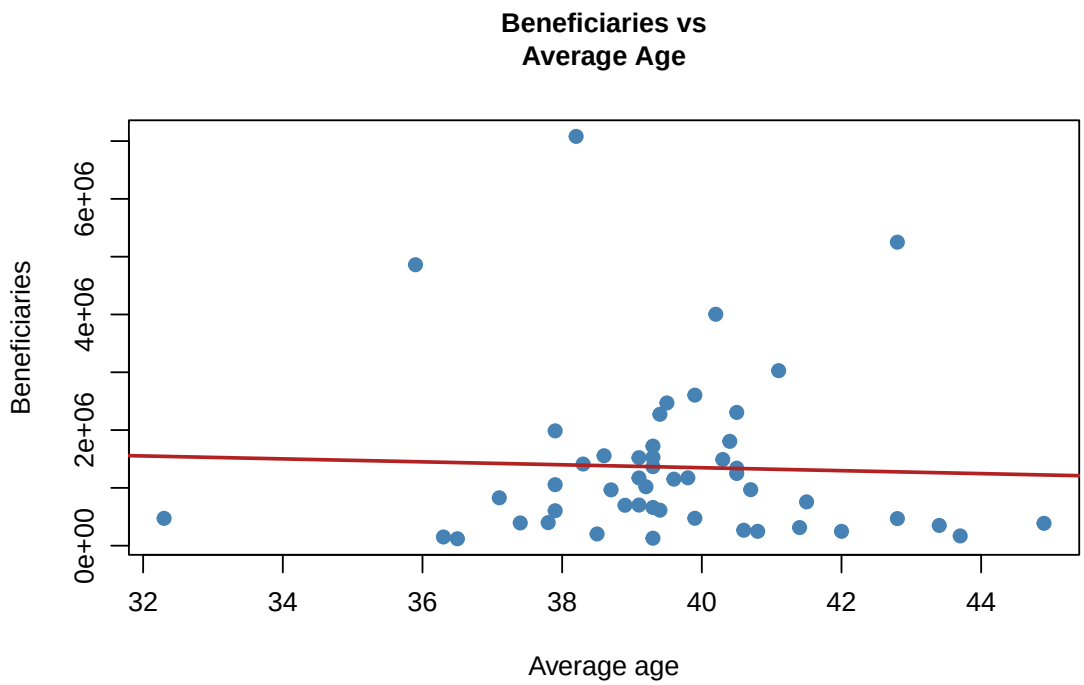
Appendix C: Exploratory Data Analysis

Response Variable Plot

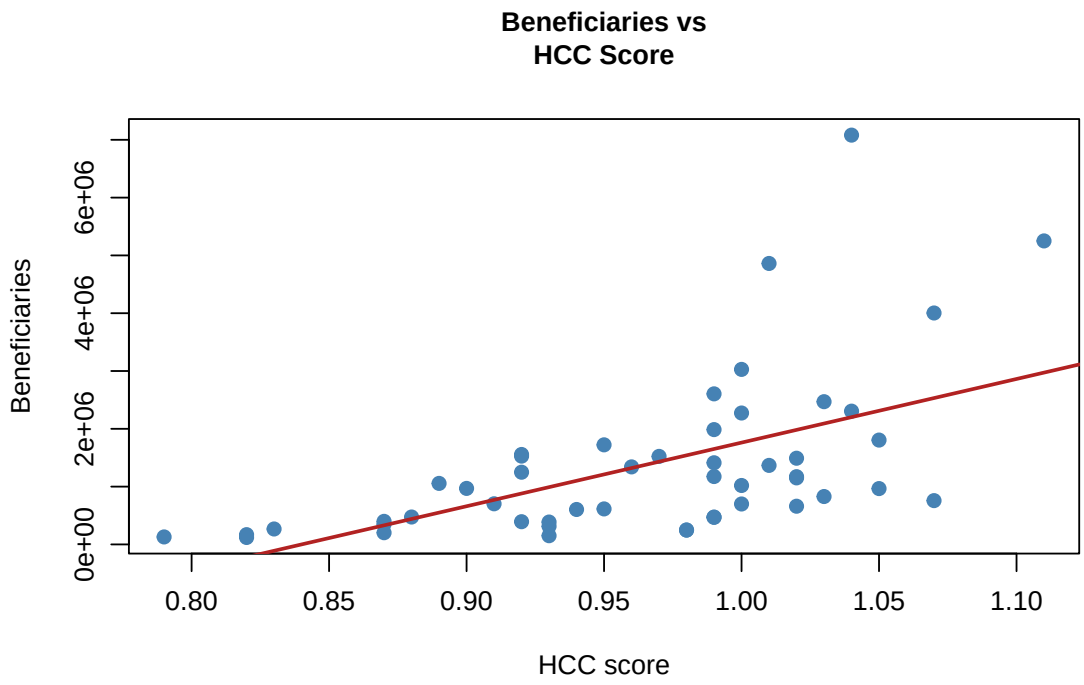


This histogram shows a strong right-skew, with most states clustered at lower beneficiary counts and a small number of very large states extending the right tail. This shape supports considering `log(Y)` in later regression work.

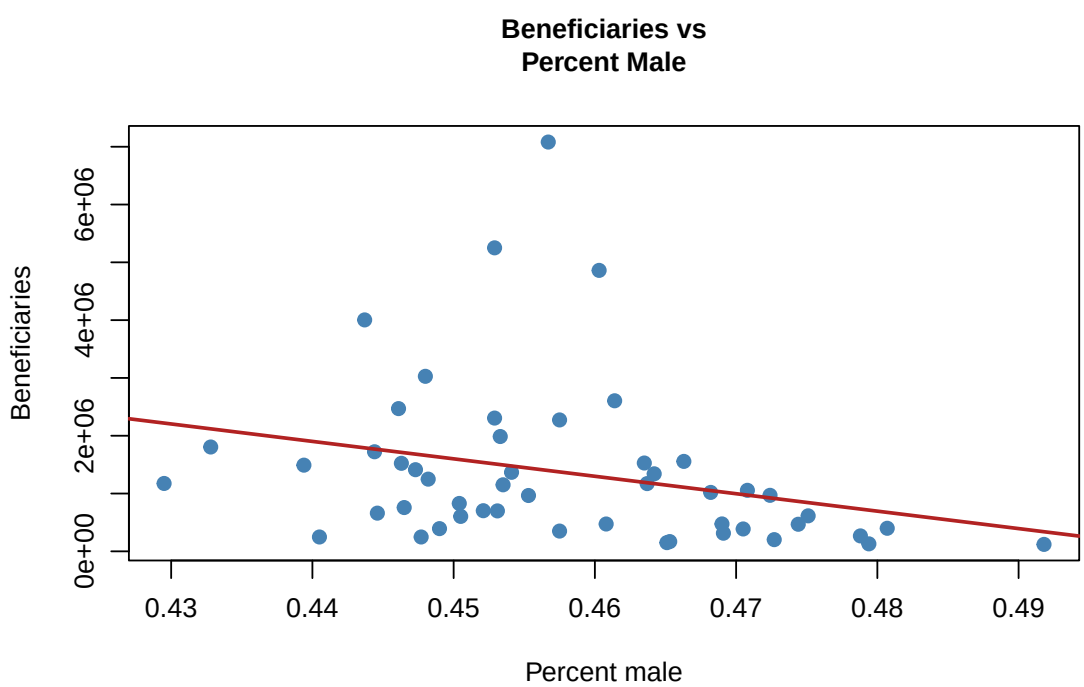
Quantitative Predictor Plots



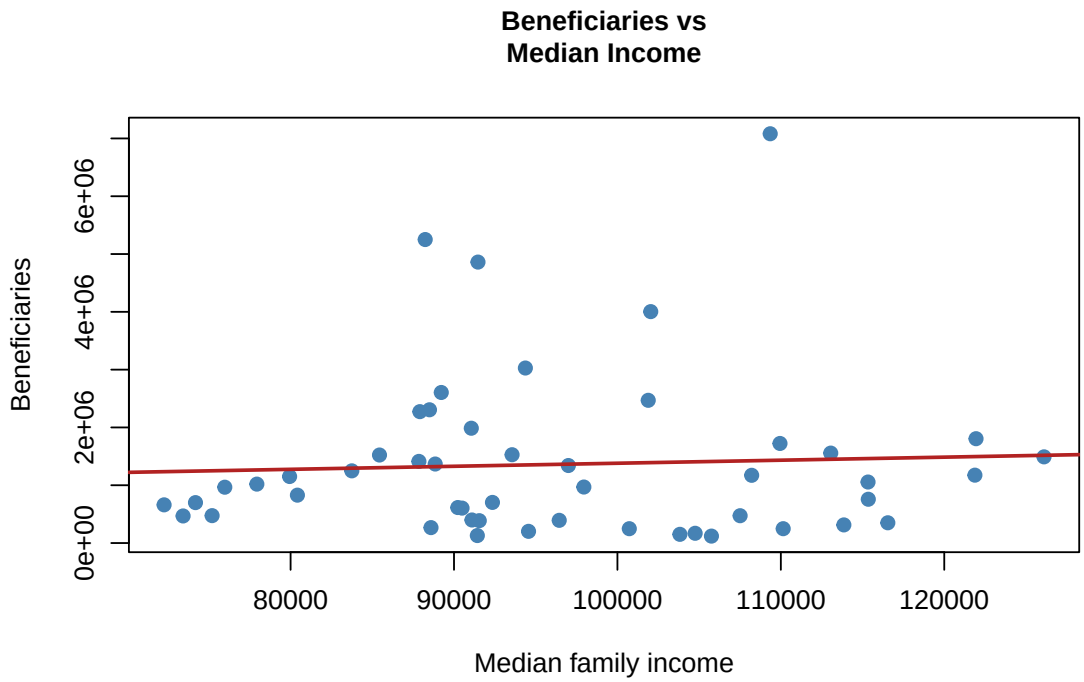
This scatterplot shows only a weak overall linear relationship between average age and beneficiary totals, which matches the small raw correlation of -0.038. The broad vertical spread suggests age may matter more through interactions than as a stand-alone predictor.



This plot shows the clearest positive quantitative pattern in the EDA. States with higher HCC scores tend to have larger beneficiary totals, consistent with the positive raw-response correlation of 0.562.

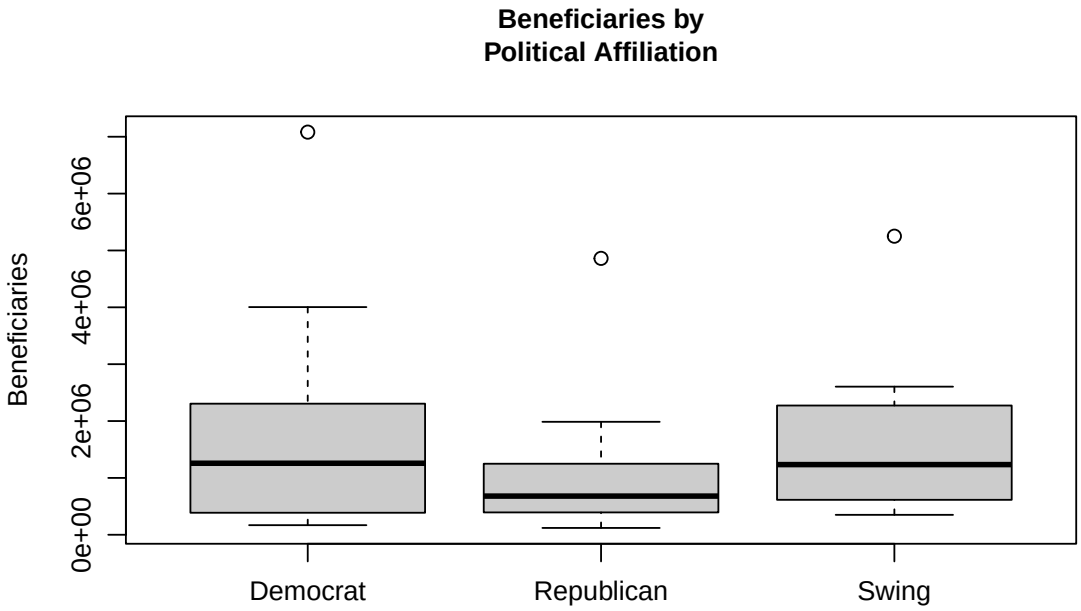


This scatterplot suggests a negative association between percent male and beneficiary totals. The downward fitted line agrees with the negative correlation of -0.28, although there is still substantial variation around the trend.

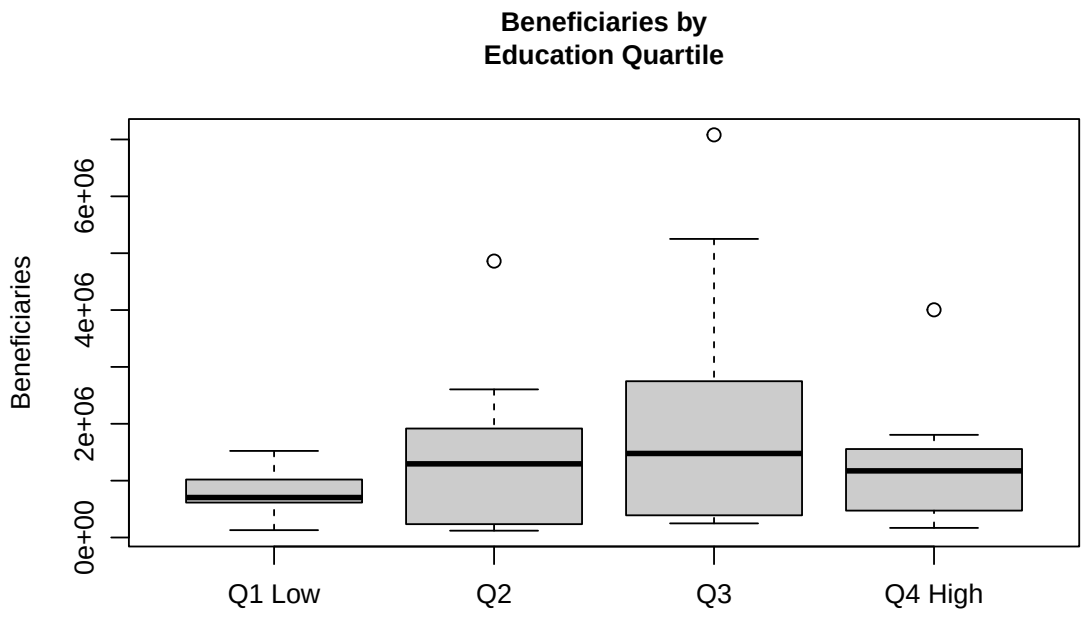


This plot shows a negligible linear relationship between median family income and beneficiary totals. The nearly flat fitted line is consistent with the very small correlation of 0.051.

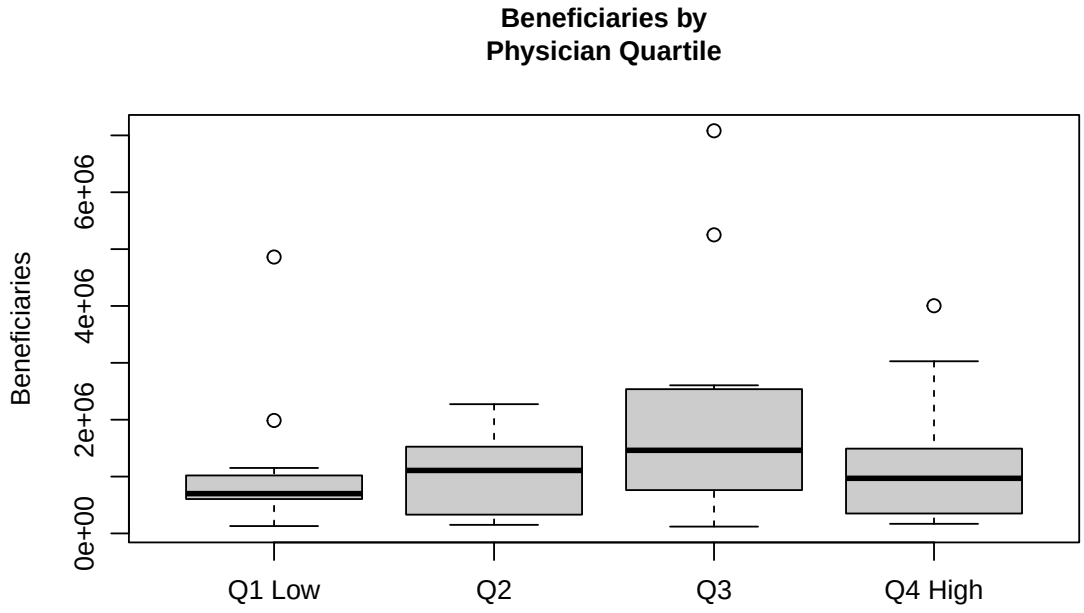
Qualitative Predictor Plots



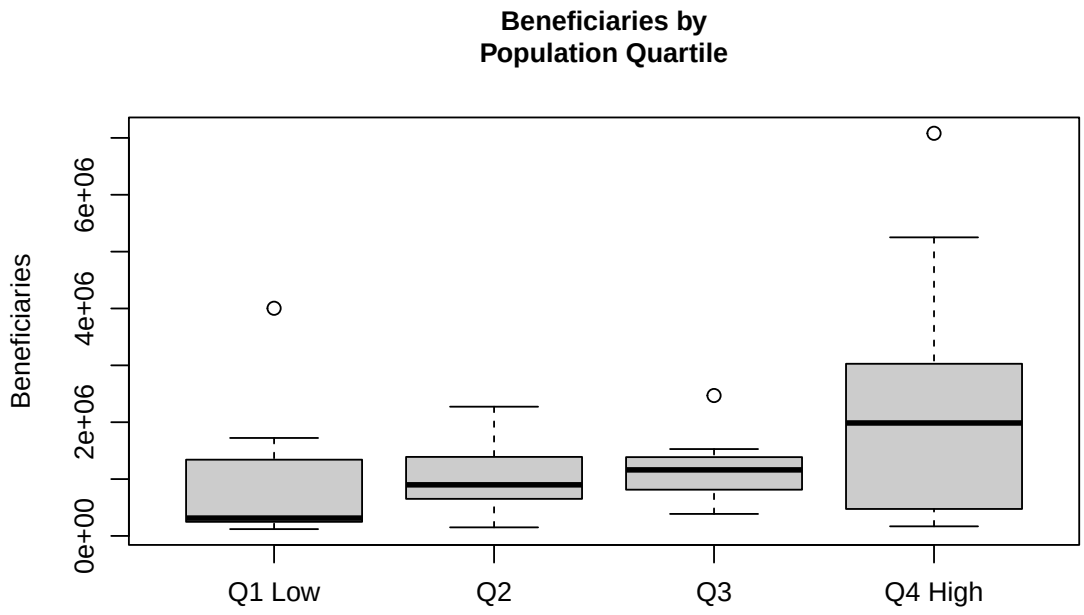
This boxplot compares beneficiary totals across political-affiliation groups. Democratic and swing states have higher average totals than Republican states in this sample, although there is still overlap across the three groups.



This plot suggests some differences across education quartiles, but the pattern is not perfectly monotone. Mean beneficiary totals rise from 819,413 in the lowest quartile to 2,083,766 in Q3 before dropping in Q4.

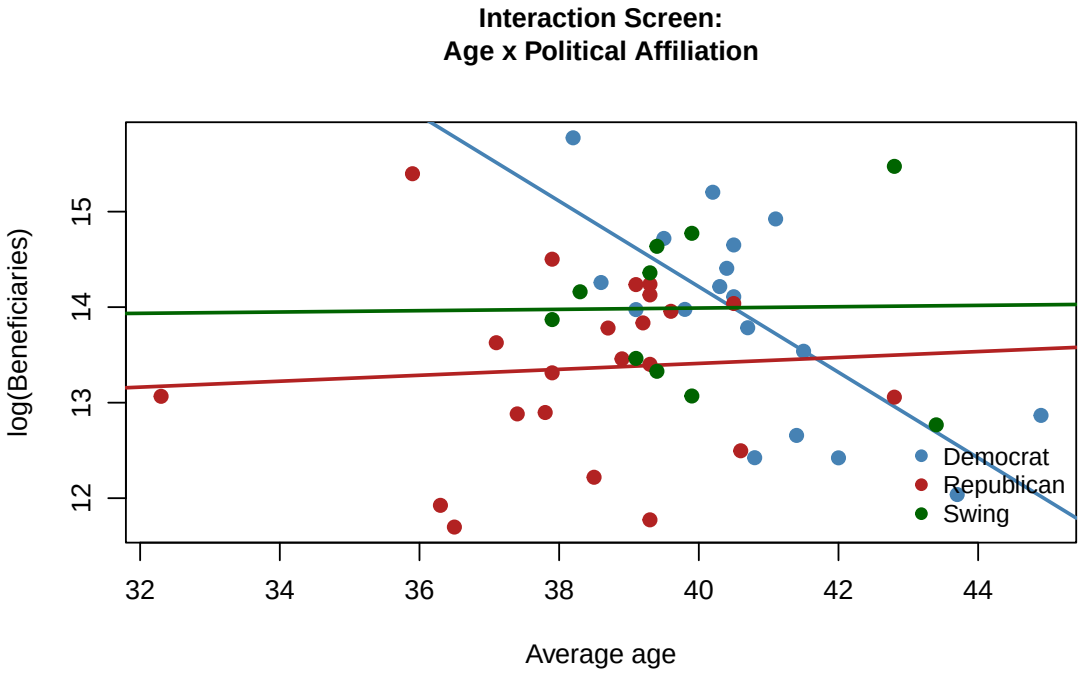


This boxplot indicates that physician availability does not increase in a simple step-by-step way with beneficiary totals. The group means vary across quartiles, but the ordering is uneven, so regression adjustment will be more informative than the unadjusted comparison alone.

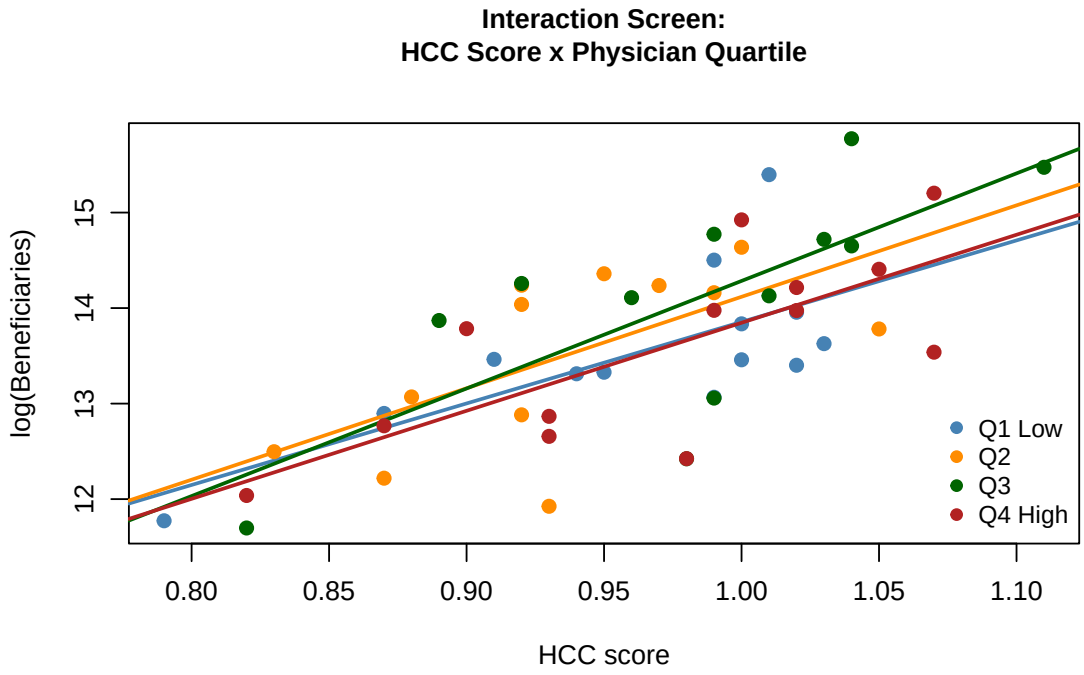


This plot shows the strongest qualitative separation in the appendix. States in higher population quartiles generally have larger beneficiary totals, with the highest-quartile mean of 2,360,061 compared with 860,492 in the lowest quartile.

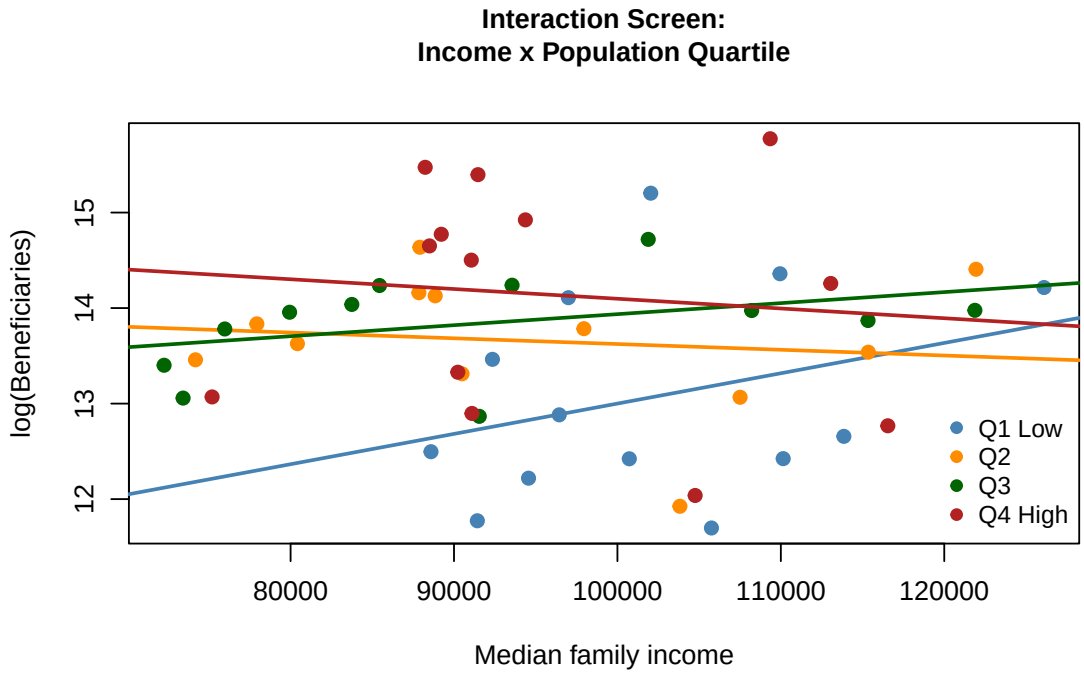
Interaction Plots



This interaction plot reveals that the relationship between a state's average Medicare age and its total number of beneficiaries depends heavily on its political affiliation. While Republican and Swing states exhibit nearly parallel, relatively flat trend lines indicating that average age has little impact on total enrollment in those regions, Democratic states display a steep downward slope. This distinct difference in slopes suggests a strong interaction effect, meaning our final model should account for political affiliation rather than assuming a uniform relationship between age and beneficiaries across all states.

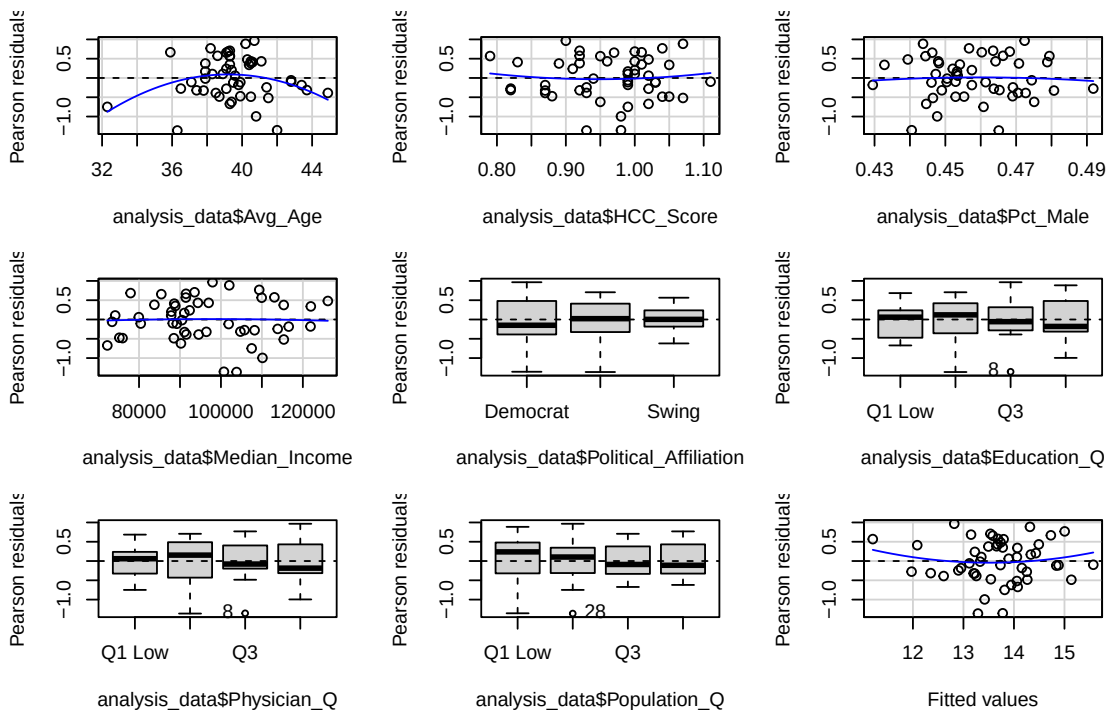


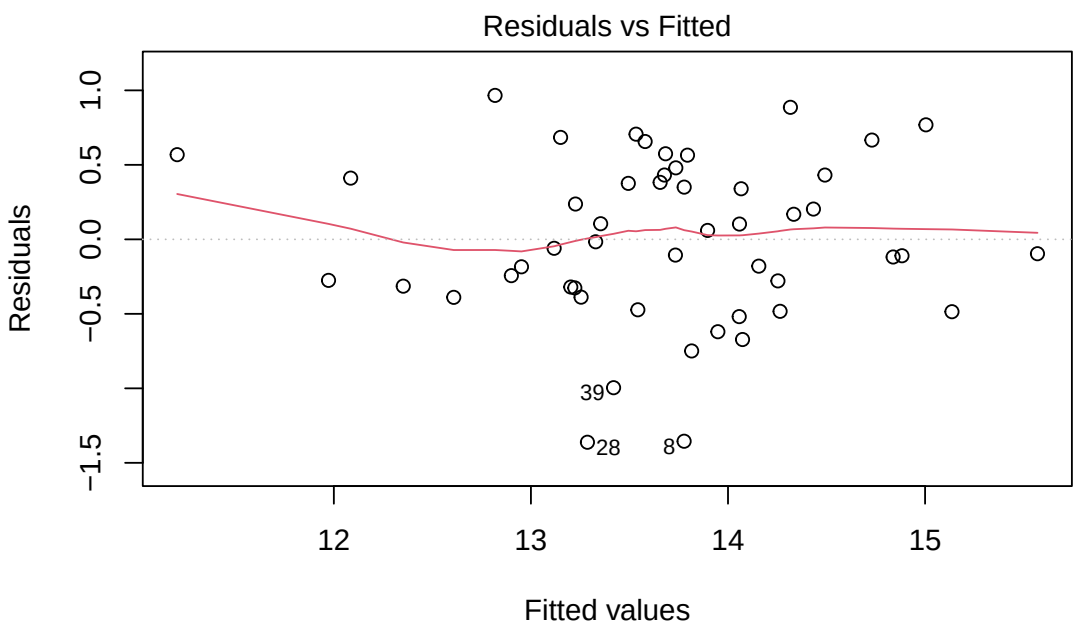
This plot examines whether physician availability modifies the relationship between a state's HCC score and its total beneficiaries. Because the trend lines for all four quartiles are heavily overlapping and nearly parallel, there is no visual evidence of an interaction effect. This suggests that the relationship between a state's medical risk and its beneficiary count remains consistent regardless of the state's physician rate, thus indicating no interaction amongst these two explanatory variables.



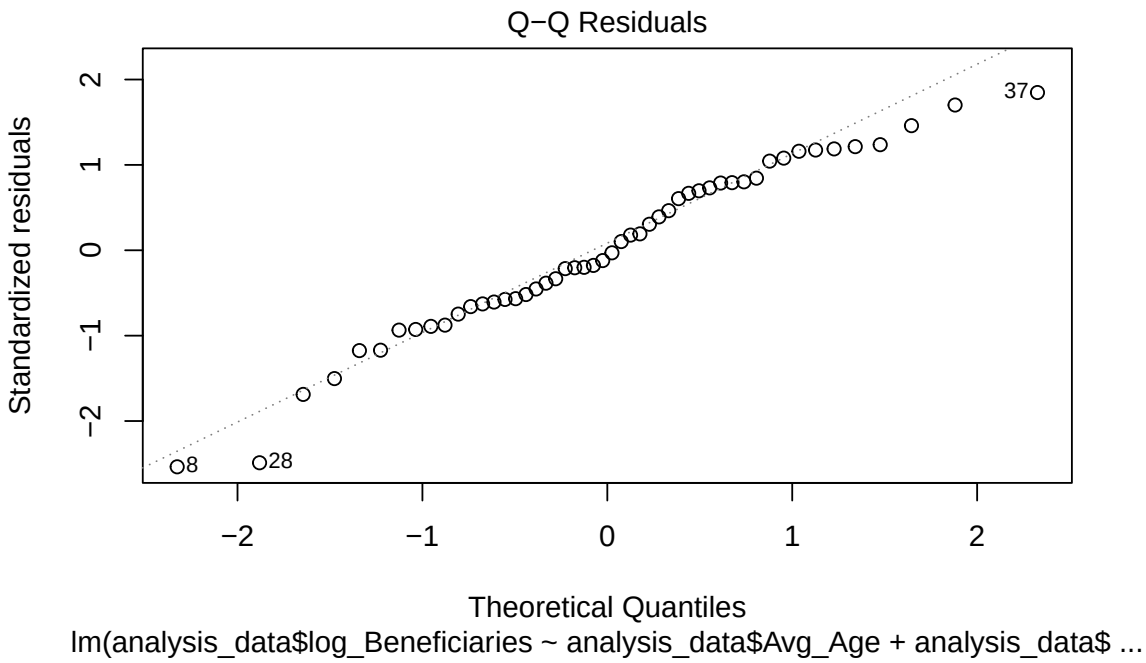
This plot examines whether a state's population size modifies the relationship between its median income and total number of beneficiaries. Because the trend lines for the four population quartiles are roughly parallel, there is no strong visual evidence of an interaction effect altering the slopes. However, it is seen that Q1 and Q3 of population sizes have flat/slightly positive slopes, while Q2 and Q4 are slightly negative, but these differences are not major enough, subjectively, to indicate any major interaction occurring.

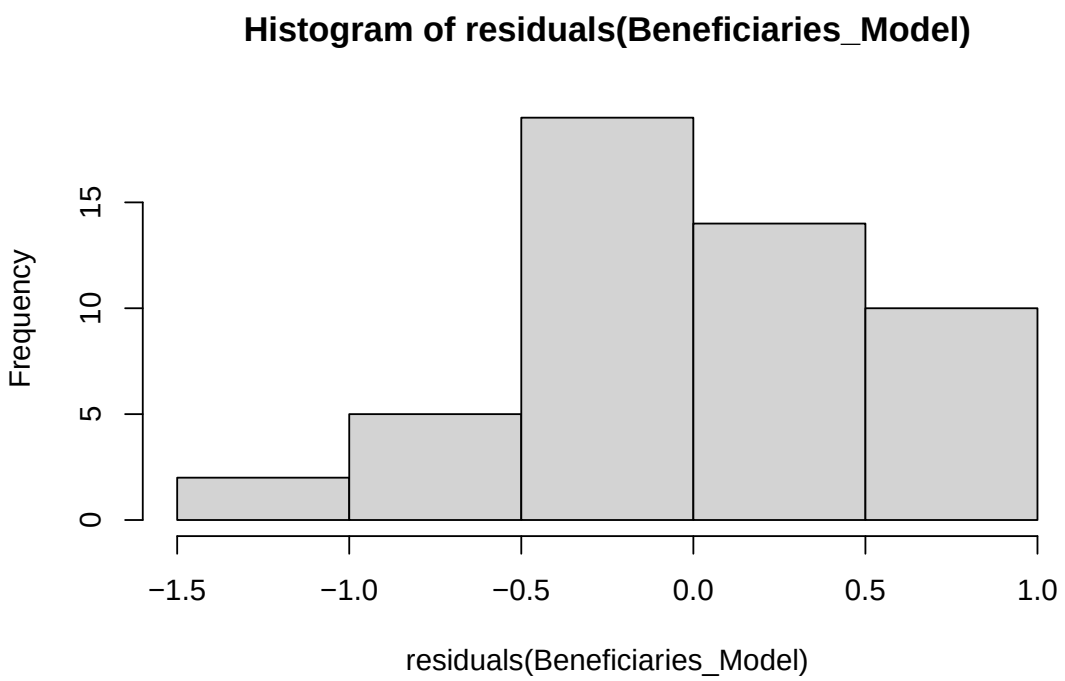
Appendix D: Final Model Output and Plots (ADD PLOTS OF ANALYSIS TO THIS SECTION)





lm(analysis_data$log_Beneficiaries ~ analysis_data$Avg_Age + analysis_data$ ...


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(analysis_data$log_Beneficiaries ~ analysis_data$Avg_Age + analysis_data$ ...
37
8
28
−2
−1
0
1
2


Histogram of residuals(Beneficiaries_Model)
Frequency
residuals(Beneficiaries_Model)
0
5
10
15
−1.5
−1.0
−0.5
0.0
0.5
1.0

Appendix E: References

- Age and sex. (2023). American Community Survey. https://data.census.gov/chart/ACSST1Y2023.S0101?q=median+age+by+state&measure=LINE32&sort=abc
- Electoral map: Blue or red states since 2000. (2026). Electoral Map: Blue or Red States since 2000. 270towin.
- Medicare Geographic Variation - by National, State & County. (2025). Medicare Geographic Variation - by National, State & County. Centers for Medicare & Medicaid Services. https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-geographic-comparisons/medicare-geographic-variation-by-national-state-county
- Newitt, P. (2025). Physicians per capita in all 50 states. Becker's ASC Review. https://www.beckersasc.com/asc-news/physicians-per-capita-in-all-50-states-2/
- US Census Bureau. (September 15, 2024). Median annual family income in the United States in 2023, by state (in U.S. dollars) [Graph]. In Statista. Retrieved April 12, 2026, from https://www.statista.com/statistics/236804/median-family-income-in-the-united-states-by-state/
- Erickson, W., Lee, C., von Schrader, S. (2026). Disability Statistics from the American Community Survey (ACS). Ithaca, NY: Cornell University Yang-Tan Institute (YTI). Retrieved from Cornell University Disability Statistics website: www.disabilitystatistics.org

Note: Some of the vocabulary, grammar, and other wording in this report was refined using ChatGPT to give the most accurate data descriptions and analysis